Chemical and Electrical Burn

TB-02 | Trauma & Burns

Quick flow

01 Scene / PPE [EMT / AEMT / Paramedic]

Identify agent or electrical source; do not enter until hazards are controlled.

02 Stop Exposure [EMT / AEMT / Paramedic]

Disconnect power when safe • remove contaminated clothing and jewelry • brush off dry chemical.

03 Decontaminate [EMT / AEMT / Paramedic]

Brush away dry particulate before irrigation. For most chemical skin/eye exposures, begin prompt copious irrigation without delaying ordinary eye irrigation for product identification. Known water-reactive contamination requires immediate HazMat/Poison Help direction; open TE-09 for agent-specific exceptions. Protect unaffected skin and responders.

04 Airway or Trauma Threat? [EMT / AEMT / Paramedic]

Treat inhalation injury, dysrhythmia, fall, blast, or multisystem trauma.

05 Monitor + Treat [AEMT / Paramedic]

For high-voltage/lightning exposure, loss of consciousness, chest pain, dysrhythmia, abnormal ECG, or significant trauma: cardiac monitoring, vascular access, burn care, pain control, and reassessment.

06 Burn-Center Transport [EMT / AEMT / Paramedic]

Consult a burn center for every chemical injury, high-voltage (≥1,000 V) injury, or lightning injury. Arrange consultation and consider follow-up after low-voltage injury, including delayed symptoms or visual complaints; use the regional destination plan and notify early.

Overview

- Stop the exposure while protecting responders, then identify chemical, electrical, lightning, or associated traumatic injury.
- Chemical and electrical burns may cause severe internal injury despite limited external findings.

Provider-Level Actions

Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT

- Protect the crew, stop exposure, remove contaminated clothing and jewelry, and brush off dry particulate before irrigation. For most chemical skin/eye exposures, begin prompt copious irrigation; do not delay ordinary chemical-eye irrigation while identifying an unknown agent. Remove contact lenses when easily possible. For known water-reactive contamination, follow immediate HazMat/Poison Help direction and TE-09.
- After decontamination, cover burns with clean dry dressings except when white phosphorus is suspected: keep affected areas wet with cool water or wet dressings to prevent reignition until particles are removed. Protect the rest of the patient from hypothermia, manage associated trauma, and transport promptly.

AEMT

- Perform all EMT care plus vascular access, reassessed fluid only for shock or significant burn-resuscitation need, and analgesia within standing orders.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.
- Prioritize cardiac monitoring after high-voltage/lightning exposure, loss of consciousness, chest pain, dysrhythmia, abnormal ECG, or significant associated trauma; do not delay transport.

Paramedic

- Perform all prior care plus advanced airway treatment, dysrhythmia management, and treatment of shock or compartment-threatening injury.
- Consult Poison Help or Medical Control for agent-specific care and a burn center for every chemical injury, high-voltage (≥1,000 V) injury, or lightning injury. Arrange burn-center consultation and consider follow-up after low-voltage injury, including delayed symptoms or visual complaints.

Indications

- Patient with suspected chemical and electrical burn requiring prehospital assessment or treatment.

Warnings / precautions

- Do not neutralize a chemical on the patient unless specifically directed by Poison Control, Medical Control, or HazMat; chemical reactions may generate heat.
- Do not apply ice, creams, ointments, or routine wet sheets to significant burns. White phosphorus is an exception: keep the affected area wet until particles are removed while keeping uninjured areas warm.

Clinical pearls

- Apply the trauma field-triage and destination plan: high-risk patients to the highest-level appropriate trauma center within regional constraints; favor pediatric-capable trauma care for children. Use a closer stabilization hospital only for an immediate critical need that cannot safely be managed during longer transport, under the destination plan/Medical Control. Notify early, document the reason, and activate the predefined transfer process. Do not delay ground movement solely while awaiting aircraft.
- Electrical current may cause dysrhythmia, deep tissue injury, rhabdomyolysis, fractures, or traumatic falls.
- Assess entrance and exit wounds, distal perfusion, neurologic status, and circumferential injury.

Related pathways / contacts

TE-09 Chemical / HazMat Exposure - Agent-specific decontamination, water-reactive contamination, white phosphorus, and antidote restrictions.

TB-11 Eye Trauma - Ordinary chemical-eye irrigation and known or strongly suspected open-globe exception.

TB-09 Thermal Burn - Burn assessment, controlled fluid rates, consultation, and hypothermia prevention.

Source and references

Current Tennessee EMS scope of practice, verified individual credentials, and Claiborne County standing orders control provider authorization.

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